Stefano Lemerle

Computer Engineering MSc student, ML for audio and music

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Education

2025–present **MSc Computer Engineering (AI and data analytics)**, *Politecnico di Torino*, Turin, Italy, 29.6/30
Exchange semester at KAIST (fall 2026), funded by a Politecnico di Torino Extra-EU mobility scholarship. Courses: Kernel System Security, Embedded Operating Systems, Intelligent Robotics, Theory of Programming Languages.

2022–2025 **BSc Computer Engineering**, *Politecnico di Torino*, Turin, Italy, 108/110

Selected coursework (grade out of 30, L = cum laude)

ML and data Machine learning and pattern recognition (30L), Big data processing and analytics (30), Data science and database technology (30L), GeoAI: geospatial artificial intelligence (30)

Signals Signal theory and signal processing (23), Automatic control (30)

Math Mathematical analysis II (29), Mathematical methods for engineers: probability, statistics, distribution theory (28), Linear algebra and geometry (25)

Systems Operating systems (30L), Computer architectures (30), System and device programming (30L), Software engineering (30)

Projects

2026 **Chorus Parameter Estimation**, *Python, PyTorch, Rust* repo
Trained a small CNN to recover the four knob settings of my chorus plugin from the processed audio alone, using 26,000 two-second clips. Feeding it mid/side instead of left/right channels cut the validation error from 0.060 to 0.033; it beats a cross-correlation baseline that also hears the dry signal on rate, depth and mix, but not on delay.

2026 **CrazyChorus**, *Rust, Truce, Slint* repo
Wrote a stereo chorus plugin (AU, VST3, CLAP, standalone) with hand-written DSP: modulated delay lines with cubic Hermite interpolation and offset stereo LFOs. Its DSP crate is reused in the estimation project.

2026 **OS161 virtual memory**, *C, OS161* repo
Built demand paging, FIFO page replacement with swap and a coremap; added fork, exit, waitpid and sbrk.

2025 **LandTiger Tetris**, *C, LPC1768* repo
Implemented joystick and button controls, rotation, soft and hard drop, and a saved high score.

2026 **Race the Guild**, *React, Express, SQLite* repo
Built a timed, Dune-themed route-planning game: React client, Express REST API, Passport login, leaderboard.

Skills

Programming Python, C, Rust, Java, JavaScript, SQL, MATLAB, OCaml, Lean, Bash; assembly (MIPS, ARM, RISC-V)

ML and data PyTorch, NumPy, pandas, scikit-learn, Matplotlib; PySpark, Hadoop

DSP and audio Filters, FFT/STFT, fractional delay lines, LFOs, mid/side; real-time plugins (Truce, Slint); Logic Pro

Systems Linux, Git, kernel programming (OS161), multithreaded programming, bare-metal ARM Cortex-M3

Web React, Node.js, Express, REST APIs, SQLite, Passport

Languages English (native; IELTS 8.5, CEFR C2), Italian (native), French (beginner), Spanish (beginner)

Music and audio

Produce, mix and master music since 2019, some released on streaming platforms. Built a home studio in 2026, treating a square room with 35 acoustic panels. Play piano and guitar; music theory.

Other experience

2025–2026 **Private tutoring**, *high-school maths, physics, chemistry and computer science*

2023–2025 **Italian–English interpreter**, *guided tours and olfactory experiences, Cherasco*

2024 **Website for a B&B**, *La Mela Antica, Cherasco*, Italian and English, lamelaantica.com